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59. You are given normal boiling points and standard enthalpies of vapourisation. Calculate the entropy of vapourisation of liquids listed below.

S. No	Liquid	Boiling points (^{0}C)	ΔH (kJ mol^{-1})
1.	Ethanol	78.4	+ 42.4
2.	Toluene	110.6	+ 35.2

60. For the reaction $Ag_2O(s) \rightarrow 2Ag(s) + \frac{1}{2} O_2(g)$: $\Delta H = 30.56$ kJ mol^{-1} and $\Delta S = 6.66 JK^{-1}mol^{-1}$(at 1 atm). Calculate the temperature at which ΔG is equal to zero. Also predict the direction of the reaction (i) at this temperature and (ii) below this temperature.

61. What is the equilibrium constant Keq for the following reaction at 400K.

$2NOCl(g) \rightleftharpoons 2NO(g) + Cl_2(g)$,

given that $\Delta H^0 = 77.2$kJ mol^{-1} ;

and $\Delta S^0 = 122 JK^{-1}mol^{-1}$.

62. Cyanamide (NH_2CN) is completely burnt in excess oxygen in a bomb calorimeter, ΔU was found to be −742.4 kJ mol−1, calculate the enthalpy change of the reaction at 298K. $NH_2CN(s) + \frac{3}{2} O_2(g) \rightarrow N_2(g) + CO_2(g) + H_2O(l)$ ΔH= ?

63. Calculate the enthalpy of hydrogenation of ethylene from the following data. Bond energies of C – H, C – C , C = C and H – H are 414, 347, 618 and 435 kJ mol−1.

64. Calculate the lattice energy of $CaCl_2$ from the given data

$Ca(s) + Cl_2(g) \rightarrow CaCl_2(s)$ $\Delta H_f^0 = -795$ kJ mol^{-1}

Atomisation : $Ca(s) \rightarrow Ca(g)$ $\Delta H_1^0 = +121$ kJ mol^{-1}

Ionisation : $Ca(g) \rightarrow Ca^{2+}(g) + 2e^-$ $\Delta H_2^0 = +2422$ kJ mol^{-1}

Dissociation : $Cl_2(g) \rightarrow 2Cl(g)$ $\Delta H_3^0 = +242.8$ kJ mol^{-1}

Electron affinity : $Cl(g) + e^- \rightarrow Cl^-(g)$ $\Delta H_4^0 = -355$ kJ mol^{-1}

65. Calculate the enthalpy change for the reaction $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$ from the following data.

$2Fe + \frac{3}{2} O_2 \rightarrow Fe_2O_3$; $\Delta H = -741$ kJ

$C + \frac{1}{2} O_2 \rightarrow CO$; $\Delta H = -137$ kJ

$C + O_2 \rightarrow CO_2$; $\Delta H = -394.5$ kJ